**Panskura Banamali College (Autonomous)**

**B.Sc.(H) 6th Semester Examination -2022 (Online Mode)**

**Subject: Computer Science** **Paper: CC14T**

**Paper Name: Computer Graphics**

**F.M.: 40** **Time : 2 Hours**

---

**Answer any four questions:** **4x10=40**

1. Explain Sutherland Hodgman polygon clipping algorithm. 10

2. Rasterize the line from (-1,1) to (5,-8) using Bresenham's line drawing Algorithm. 10

3. Find the reflection of a triangle defined by the vertices A(1,1), B(5,1) and C(1,5) about a line y=2x+10. 10

4. Prove that rotation followed by translation is not same as translation followed by rotation in three dimension. 10

5. Scale the surface A(2, 2, 2), B(4, 4, 4), C(5, 5, 5), D(6, 6, 6) with respect to point (7, 7). 10

6. Consider a Non-Interlaced raster system with resolution of 1280 By 1024, a refresh rate of 60 Hz, a horizontal retrace time of 5 Microseconds and a vertical retrace time of 500 μs. What is the fraction of the total refresh time per frame spent in horizontal retrace of the electron beam? 10

***